

ASSIGNMENT : 7

TITLE : Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement : Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives : To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory : Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Exercise :

1. Create a **Shape application** where Circle and Rectangle inherit from Shape and calculate area.

```
class Shape {
    void area() {
        System.out.println("Area of Shape");
    }
}

class Circle extends Shape {
    double radius = 7;

    void area() {
        System.out.println("Area of Circle = " + (3.14 * radius * radius));
    }
}

class Rectangle extends Shape {
    int length = 15;
    int breadth = 5;

    void area() {
        System.out.println("Area of Rectangle = " + (length * breadth));
    }
}

public class Main {
    public static void main(String[] args) {
        Circle c = new Circle();
        Rectangle r = new Rectangle();

        c.area();
        r.area();
    }
}
```

Output

Generated files

```
Area of Circle = 153.86
Area of Rectangle = 75
```

2. Develop a **E-Commerce Product System** where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```
interface Product {
    void display();
}

class ProductDetails {
    String name;
    double price;

    ProductDetails(String name, double price) {
        this.name = name;
        this.price = price;
    }
}

class Electronic extends ProductDetails implements Product {
    Electronic(String name, double price) {
        super(name, price);
    }

    public void display() {
        System.out.println("Electronic: " + name + " Price: ₹" + price);
    }
}

class Clothing extends ProductDetails implements Product {
    Clothing(String name, double price) {
        super(name, price);
    }

    public void display() {
        System.out.println("Clothing: " + name + " Price: ₹" + price);
    }
}

class Grocery extends ProductDetails implements Product {
    Grocery(String name, double price) {
        super(name, price);
    }

    public void display() {
        System.out.println("Grocery: " + name + " Price: ₹" + price);
    }
}
```

```
public class Main {  
    public static void main(String[] args) {  
  
        Electronic e = new Electronic("Laptop", 65000);  
        Clothing c = new Clothing("Shirt", 1200);  
        Grocery g = new Grocery("Rice", 900);  
  
        e.display();  
        c.display();  
        g.display();  
    }  
}
```

Output

Generated files

```
Electronic: Laptop Price: ₹65000.0  
Clothing: Shirt Price: ₹1200.0  
Grocery: Rice Price: ₹900.0
```

Outcome :

- Understood inheritance and interfaces in Java.
- Implemented code reusability using **extends**.
- Used interfaces to achieve abstraction.

Conclusion :

This assignment demonstrated inheritance and interfaces in Java. Inheritance promotes code reuse, while interfaces provide abstraction and multiple inheritance of type, resulting in modular and maintainable applications.